
The Chain of Thought Is Working Memory, Not Overflow

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Abstract

A language model’s internal workspace holds at most a few steps of chained reasoning state, at every scale we test, and the chain of thought holds the rest. A common reading of J-space, the concept workspace a fitted Jacobian lens reads from the residual stream, is that reasoning is carried internally and writing relieves the workspace by externalizing state it would otherwise hold. Three measurements contradict the relief picture. Instructed to answer chained arithmetic without writing, models from 1.5B to 671B parameters collapse beyond one to two dependent steps; a 671B model, so instructed, fails two chained additions. Models write every intermediate value from the first step at every difficulty, so externalization begins at no capacity threshold. The allocation between the two sides never moves: damaging the workspace does not increase writing, capping writing does not move computation inward, and ablating the model’s active J-space concepts at the strongest dose that preserves ordinary generation changes nothing, even at the one-to-two-step depths where the workspace does perform the arithmetic. We then show how this memory works, with causal interventions on six unmodified models (used as released, with no fine-tuning or task-specific training) from four families on arithmetic and narrative state-tracking tasks. Each written value is held at its own token position as a register: overwriting the hidden state there with the state a different number would have produced, text unchanged, redirects the final answer to that never-written number in 0.70 to 1.00 of cases, while size-matched random noise redirects at most 0.01. The register is read through attention to its position (blocking it collapses dependence to at most 0.007 with matched controls unchanged), the read completes by two thirds of the network’s depth, and two positions act as independent stores whose planted values compose into a sum appearing nowhere in the input (0.88 to 0.90). For oversight, identical visible text can carry different stored values with different downstream behavior, so a chain of thought exposes the symbols a computation operates on without guaranteeing the hidden content behind them.

1 Introduction

The chain of thought is the primary working memory of serial reasoning in language models; the internal workspace behind it holds at most a few dependent steps, and writing is not overflow from it. We establish this claim with causal interventions on six unmodified models from four families (Qwen, Phi, OLMo, DeepSeek-R1-Distill), on two task families with exact ground truth for every intermediate value.

The capacity half addresses the overflow picture of externalization: reasoning state is carried internally, and writing takes over when the workspace can no longer hold it. This was our pre-registered hypothesis, and it is a natural reading of workspace accounts of reasoning, including J-space, the

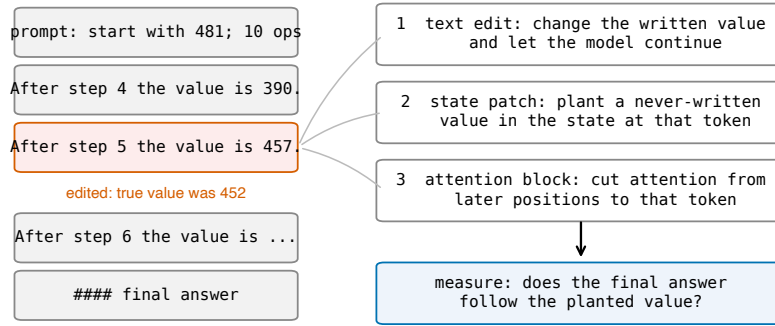


Figure 1: The edit-and-continue protocol. A correct worked solution is truncated after one edited value line and the model continues. Three interventions target the edited value: changing the visible token, overwriting the hidden state at that position with the state a different value would have produced, and blocking attention from later positions to that position. The readout is whether the final answer follows the planted value.

38 concept workspace a fitted Jacobian lens reads from the residual stream [Gurnee et al., 2026]. A
39 workspace that writing relieves should show up in three places, and it shows up in none of them.

- 40 1. **No fallback.** Instructed to answer chained arithmetic with no writing (token counts confirm
41 the channel is closed), models collapse beyond one to two dependent steps; DeepSeek V3.2,
42 at 671B parameters, answers 0.23 at one step and near zero above (Section 3).
- 43 2. **No threshold.** Models write 0.93 to 1.00 of ground-truth intermediates from one-step
44 problems onward, at every scale from 1.5B to 671B (Section 3).
- 45 3. **No reallocation.** Damaging the workspace does not increase writing, capping writing does
46 not move computation inward, and ablating the top active J-space concepts at a generation-
47 preserving dose changes neither accuracy nor what gets written (Section 3).

48 The second half is the mechanism of the memory that remains.

- 49 4. **Registers.** The hidden state at a written value’s position is a settable value register: planting
50 a never-written value there, text unchanged, redirects the answer at 0.70 to 1.00 across all
51 six models; restoring the correct state under corrupted text reverts it at 0.88 to 1.00; size-
52 matched random noise redirects at most 0.01 on instruction-tuned models (Section 4).
- 53 5. **Retrieval.** The register is read through attention to its position: blocking that attention
54 collapses dependence to at most 0.007 on four models in three families, with neighbor,
55 random, and earlier-value blocks at the unblocked level and fluency unchanged. Layer
56 sweeps time the read: single-layer patches succeed at any depth up to two thirds of the
57 network and fail after, and blocking attention only at the middle third reproduces the full
58 collapse (Section 5).
- 59 6. **Composition, and when the register is trusted.** Two positions are independently address-
60 able, composing planted values into their sum (0.88 to 0.90). Models read the register by
61 default; which prompt evidence triggers recomputation is model-specific, and reliance on
62 the trace tracks whether the model can regenerate the state, which reconciles conflicting
63 reports on how trace reliance scales (Sections 4 and 6).

64 Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal,
65 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026]
66 and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether
67 an internal alternative exists, whether the behavior is native to unmodified pretrained models, how
68 the value is retrieved, or whether stores compose.

69 2 Tasks and protocol

70 Both task families give exact ground truth for every intermediate value, so the downstream effect of
71 any edit is computable. Arithmetic chains start from a three-digit number and apply d signed two-
72 digit additions and subtractions. Narrative state-tracking problems describe a character whose count
73 of an item changes through varied events, with varied names, items, verbs, and sentence forms. In
74 both, the model writes one line per step giving the running value.

75 The protocol takes a correct worked solution, changes the value at one step j , truncates immedi-
76 ately after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures
77 the sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five
78 samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap
79 intervals over items; the effects that carry the argument exceed their controls by far larger mar-
80 gins. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-7B-
81 Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; frontier models (DeepSeek V3.2, Claude Son-
82 net 4.5, Llama 3.3 70B) are tested behaviorally through assistant prefill. Model revisions, prompts,
83 layer bands, and configurations are in the appendix; code and data will be released.

84 3 The internal workspace holds a few steps at most

85 Three measurements bound the internal side of serial reasoning. First, capacity: instructed to output
86 only the final answer, with token counts confirming nothing else was generated, models fail beyond
87 one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at $d = 4$; DeepSeek V3.2 at 671B:
88 0.23 at $d = 1$, near zero above), and the values used are random, so nothing can be recalled rather
89 than computed. Second, the write policy: models write 0.93 to 1.00 of ground-truth intermediates
90 in free generation from $d = 1$ through $d = 64$, at every scale from 1.5B to 671B, so writing does
91 not begin at any capacity threshold. Third, reallocation: a residual-stream lesion strong enough to
92 cross the accuracy cliff makes traces disintegrate rather than expand; hard token budgets compress
93 wording up to 2.5 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve
94 tokens per step; and ablating J-space [Gurnee et al., 2026] changes nothing at the strongest dose
95 preserving ordinary generation (calibrated on neutral text: next-token agreement 0.87, perplexity
96 ratio 1.12). The J-space ablation removes the top 16 active concepts at every position and leaves
97 chain-of-thought accuracy, bare-answer accuracy, and the amount written unchanged, on synthetic
98 chains and on GSM8K, doing no more than a size-matched random-directions ablation. The null
99 holds at one to two steps, where models do perform the arithmetic internally.

100 This bounds the particular readout tested rather than the workspace as a whole: the dose is capped
101 at generation-preserving strength, the lens is one fitted artifact, and a written token’s contextual
102 representation and its J-space component are not exclusive alternatives, since the lens samples the
103 same residual states our patches overwrite. The workspace also exceeds J-space by construction:
104 attention and MLP interiors and the KV cache fall outside the lens’s domain, and within the residual
105 stream the lens spans a fitted concept dictionary rather than the full state. Our register result is an
106 existence proof of the gap: the planted value is causally decisive, sits in residual states the lens
107 samples, and survives removal of the top 16 concepts. Beyond one to two dependent steps, closing
108 the written channel reveals no internal store for writing to relieve, at any scale tested.

109 4 The written position holds a settable value register

110 Table 1 decomposes the effect of editing a written value. Restoring the correct hidden state while
111 the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting
112 a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The
113 size-matched random control sits at or below 0.01 for the four instruction-tuned models. Because
114 different planted values produce answers following those values, the value content is the causal
115 quantity, and because the planted value is a mid-chain intermediate the model never produced, the
116 patch does not inject the answer. The effect persists with task depth: on Qwen3-4B the planted value
117 is followed at 0.98, 0.97, and 0.95 for chains of 5, 20, and 40 steps. No fine-tuning, task-specific
118 training, or prompt engineering produces this behavior; it is how the pretrained models already work.

Table 1: State patch on arithmetic chains at ten steps. Rates are conditioned on items where the text edit was followed (first column), so cross-model comparison of later columns should account for that rate. Brackets are 95 percent bootstrap intervals. The two reasoning-distilled models resolve the problem after their reasoning phase, which raises their random-control reversion; in a logged sample every reverting continuation re-derives from the problem’s starting value rather than correcting in place, and re-derivation can only produce the correct answer, so the planted-value column is unaffected.

model	text edit followed	restore correct	plant new value	random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B-Instruct	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

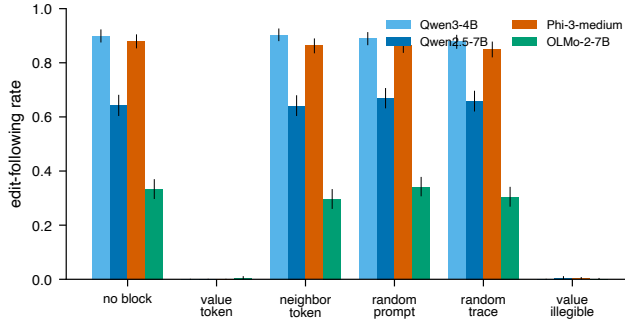


Figure 2: Rate at which the answer follows the edited value under each blocked attention target. Blocking the value’s own position collapses following on every model; matched control blocks do not. Error bars are 95 percent bootstrap intervals.

119 **The result is a property of written state, not of one template.** On the narrative task the pattern
120 replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86
121 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and
122 the random control is 0.00 in both ($n = 86$ and 89).

123 **Written positions are independently addressable.** In a task with two counters summed at the
124 end, planting a never-written value at one final position yields the corresponding mixed sum (0.94
125 and 0.93 on Qwen2.5-7B, 0.89 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90
126 and 0.88, at or above the product of the single-position rates (0.89 and 0.83). The random patch
127 leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted
128 values occur at 0.01. This is the result that separates structured memory from a strong perturbation
129 dragging the output: the reported number must be assembled at answer time from two values that
130 were set independently, at different positions, and never written anywhere.

131 5 Retrieval runs through attention to the written position

132 Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s
133 dependence on that value on every model tested (Table 2, Figure 2). Matched blocks of the adjacent
134 words, of a random prompt position, or of an earlier step’s written value leave the dependence at
135 its unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, yielding
136 answers that match neither the edit nor the correct value. The collapse holds across three families
137 and on both task formats (natural-language task: 0.730 to 0.007 on Qwen3-4B, 0.720 to 0.003 on
138 Phi-3-medium).

Table 2: Attention block on arithmetic chains: rate at which the answer follows the edited value. OLMo-2-7B continues this format less reliably (low no-block rate) and shows the same contrast.

model	no block	value position	neighbor	random position	earlier value
Qwen3-4B	0.900	0.000	0.903	0.778	—
Qwen2.5-7B-Instruct	0.643	0.000	0.640	0.670	0.658
Phi-3-medium	0.880	0.000	0.863	0.865	0.850
OLMo-2-7B	0.333	0.005	0.297	0.342	0.305

The block removes the value, not the ability to generate. Under the value-position block the unparseable-output rate stays at 0.000 (0.040 on OLMo-2-7B, equal to its unblocked rate). The model produces a fluent, parseable number that matches neither the edited value nor the correct one: it computes forward from a value it can no longer retrieve.

Retrieval occurs in the middle third of the network. Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers). Overwriting the value’s hidden state at any single layer from 0 through 22 redirects the answer at 0.90 to 0.94, matching the full multi-layer patch (0.92), while single-layer patches at layer 24 and beyond redirect at 0.02 or less ($n = 60$): the register can be rewritten anywhere upstream of its consumers and nowhere downstream of them. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged (0.857 and 0.863), and no single layer’s block matters (0.84 to 0.89 at every layer, $n = 80$): the read is distributed redundantly across the middle third’s layers and confined to it. The two sweeps agree on the same boundary, near layer 23: the value is written into the position’s residual stream from early depth, retrieved by attention across the middle third, and causally inert afterward. Coarse sweeps on two further models (appendix) show the structure generalizes while the boundary’s depth is model-specific: on Phi-3-medium, early- and mid-third patches redirect at 0.89 and 0.87 and late-third patches at 0.007, matching the Qwen3-4B shape, while on Qwen2.5-7B patches in every third redirect at 0.70 to 0.73, so its reads extend into the final third of its 28 layers.

Reading is the preferred path, not the only one. Replacing the value’s characters with unreadable dots, without touching attention, also removes following, and differently by model: the Qwen models reconstruct the value from the previous line and recover the correct answer in 0.60 to 0.73 of cases, while Phi-3-medium does not recover (0.00).

6 When the written value is read versus recomputed

Models read the register by default. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value the prompt generatively defines (the prompt states the two numbers whose sum becomes the running value) is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly, reverting most when a stated checkpoint is the value’s only source. Llama 3.3 70B follows edits at 0.97 whether or not the value is prompt-defined, despite solving the same chains from scratch, so the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. These conditions come from separate experiments; each within-experiment contrast is controlled, and we claim neither a single ordered scale across models nor that reconstruction cost governs the choice, which remains open.

The same test bounds the mechanism on natural benchmarks and bears on a disputed question. On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2, since a benchmark intermediate is often one operation from numbers still in the prompt and a stronger model recomputes it. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks, a result later work disputes; our contrasts indicate the direction depends on the state itself. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale we test. Faithfulness of the trace to the computation is a property of the state’s regenerability, and a single scale trend should not be expected.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. The memory role of written tokens in prior work is discussed in the introduction [Merrill and Sabharwal, 2024, Lanchantin et al., 2023, Zhu et al., 2025, Shih et al., 2026]. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state rather than working memory in general; code, plans, and multi-entity state are untested. The J-space result bounds one fitted lens at generation-preserving doses and does not measure the J-space content of the written positions themselves. The fine layer sweeps are established on one model; the cross-model sweeps are coarse, and on Qwen2.5-7B they do not resolve the read boundary. White-box evidence is at 4B to 14B; frontier evidence is behavioral. The read-versus-recompute results support within-experiment contrasts, and the only patch control is size-matched noise; a structured control planting a valid activation for a different quantity would further isolate value content.

9 Conclusion

Serial reasoning state outgrows a language model’s internal workspace within one to two dependent steps at every scale tested, writing begins at no threshold, and no intervention shifts state between the workspace and the trace, so the chain of thought is working memory rather than overflow. It works as compositional registers: settable by intervention, retrieved through attention in the middle layers, independently addressable, and decisive for the answer. For oversight the implication cuts both ways: the written positions are a causally used memory channel, and identical text can carry different stored values, so trace monitoring observes the channel while verifying the state behind it remains a separate problem.

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